

[I225] Statistical Signal Processing(E) Office Hour 6

1. Suppose a zero-mean wide-sense stationary random process $S(t)$ is predicted by the following linear predictor:

$$\hat{S}(t + \tau) = aS(t).$$

Assume that the autocorrelation values are

$$R_{SS}(0) = 10,$$

and

$$R_{SS}(\tau) = 6.$$

- (a) Write the prediction error.
- (b) Use the orthogonality principle to derive the optimal coefficient a .
- (c) Compute the corresponding minimum mean square error (MSE).
- (d) Explain how the prediction performance changes when $R_{SS}(\tau)$ becomes smaller.

Answer:

(a)

The prediction error is

$$e(t) = S(t + \tau) - \hat{S}(t + \tau).$$

Substituting the linear predictor gives

$$e(t) = S(t + \tau) - aS(t).$$

(b)

According to the orthogonality principle, the estimation error should be orthogonal to the observed data $S(t)$:

$$E[(S(t + \tau) - aS(t))S(t)] = 0.$$

Expanding the expectation gives

$$E[S(t + \tau)S(t)] - aE[S(t)S(t)] = 0.$$

Using the autocorrelation function, we obtain

$$R_{SS}(\tau) - aR_{SS}(0) = 0.$$

Therefore,

$$a = \frac{R_{SS}(\tau)}{R_{SS}(0)}.$$

For the given values,

$$a = \frac{6}{10} = 0.6.$$

(c)

The minimum MSE is

$$\text{MSE} = E \left[\left(S(t + \tau) - aS(t) \right)^2 \right].$$

Using the optimal coefficient, it can be written as

$$\text{MSE} = R_{SS}(0) - aR_{SS}(\tau).$$

Substituting the values gives

$$\text{MSE} = 10 - (0.6)(6) = 6.4.$$

Equivalently,

$$\text{MSE} = R_{SS}(0) - \frac{R_{SS}^2(\tau)}{R_{SS}(0)} = 10 - \frac{6^2}{10} = 6.4.$$

(d)

The value of $R_{SS}(\tau)$ measures how strongly the present state $S(t)$ is related to the future state $S(t + \tau)$. If $R_{SS}(\tau)$ becomes smaller, the optimal coefficient a also becomes smaller. This means that the present sample provides less useful information for predicting the future sample, and the prediction error becomes larger.

2. A zero-mean wide-sense stationary signal $S(t)$ has the following power spectral density:

$$S_{SS}(\omega) = \frac{4}{\omega^2 + 4}.$$

The signal is corrupted by additive white noise $v(t)$ whose power spectral density is

$$S_{vv}(\omega) = 1.$$

Assume that the signal and the noise are uncorrelated. The observed process is

$$X(t) = S(t) + v(t).$$

- (a) Derive the optimal non-causal Wiener filter $H(\omega)$.
- (b) Compute the value of $H(0)$.
- (c) Compute the minimum MSE of the Wiener filter.
- (d) Explain why high-frequency components are more strongly suppressed.

Answer:

(a)

For uncorrelated signal and noise, the non-causal Wiener filter is

$$H(\omega) = \frac{S_{SS}(\omega)}{S_{SS}(\omega) + S_{vv}(\omega)}.$$

Substituting the given spectra gives

$$H(\omega) = \frac{\frac{4}{\omega^2 + 4}}{\frac{4}{\omega^2 + 4} + 1}.$$

Simplifying the expression gives

$$H(\omega) = \frac{4}{\omega^2 + 8}.$$

(b)

At $\omega = 0$,

$$H(0) = \frac{4}{0^2 + 8} = \frac{1}{2}.$$

(c)

The minimum MSE is

$$\text{MSE} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{S_{SS}(\omega)S_{vv}(\omega)}{S_{SS}(\omega) + S_{vv}(\omega)} d\omega.$$

Using the given spectra,

$$\frac{S_{SS}(\omega)S_{vv}(\omega)}{S_{SS}(\omega) + S_{vv}(\omega)} = \frac{4}{\omega^2 + 8}.$$

Therefore,

$$\text{MSE} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{4}{\omega^2 + 8} d\omega.$$

Using

$$\int_{-\infty}^{\infty} \frac{1}{\omega^2 + a^2} d\omega = \frac{\pi}{a},$$

where $a = \sqrt{8}$, we obtain

$$\text{MSE} = \frac{1}{2\pi} \cdot \frac{4\pi}{\sqrt{8}} = \frac{1}{\sqrt{2}}.$$

(d)

The signal PSD decreases as $|\omega|$ increases, while the noise PSD remains constant. Therefore, the signal-to-noise ratio is higher in the low-frequency region and lower in the high-frequency region. The Wiener filter preserves frequencies where the signal is relatively strong and suppresses frequencies where the noise is dominant.

3. Consider the following one-step linear predictor:

$$\hat{X}[2] = -a_0X[0] - a_1X[1].$$

The autocorrelation sequence of the wide-sense stationary signal is given by

$$r[0] = 4, \quad r[1] = 2, \quad r[2] = 1.$$

- (a) Write the prediction error.
- (b) Apply the orthogonality principle to derive the Yule-Walker equations.
- (c) Express the equations in matrix form.
- (d) Solve for a_0 and a_1 .
- (e) Explain why the coefficient matrix is called a Toeplitz matrix.

Answer:

(a)

The prediction error is

$$\epsilon = X[2] - \hat{X}[2].$$

Since

$$\hat{X}[2] = -a_0X[0] - a_1X[1],$$

we have

$$\epsilon = X[2] + a_0X[0] + a_1X[1].$$

(b)

By the orthogonality principle, the prediction error must be orthogonal to the available data:

$$E[\epsilon X[0]] = 0,$$

and

$$E[\epsilon X[1]] = 0.$$

For $X[0]$,

$$E[(X[2] + a_0X[0] + a_1X[1])X[0]] = 0.$$

This gives

$$r[2] + a_0r[0] + a_1r[1] = 0.$$

For $X[1]$,

$$E[(X[2] + a_0X[0] + a_1X[1])X[1]] = 0.$$

This gives

$$r[1] + a_0r[1] + a_1r[0] = 0.$$

Substituting the given values,

$$1 + 4a_0 + 2a_1 = 0,$$

and

$$2 + 2a_0 + 4a_1 = 0.$$

(c)

The matrix form is

$$\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = - \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

(d)

Solving the equations,

$$4a_0 + 2a_1 = -1,$$

and

$$2a_0 + 4a_1 = -2.$$

The solution is

$$a_0 = 0,$$

and

$$a_1 = -\frac{1}{2}.$$

Therefore, the predictor becomes

$$\hat{X}[2] = \frac{1}{2}X[1].$$

(e)

The coefficient matrix is called a Toeplitz matrix because its elements are constant along each diagonal. This structure appears because the autocorrelation of a WSS process depends only on the time difference between samples, not on the absolute time index.

4. Consider the continuous-time signal

$$x(t) = \cos(2 \cdot \pi \cdot 400 \cdot t).$$

- (a) Determine the frequency f in Hz.
- (b) Determine the angular frequency ω in rad/s.
- (c) Write the Fourier transform $X(f)$.
- (d) Find the minimum sampling frequency required by the Nyquist-Shannon sampling theorem.
- (e) Determine whether sampling at 600 Hz can perfectly reconstruct the signal.
- (f) Explain the concept of aliasing.

Answer:

(a)

The general form of a cosine signal is

$$x(t) = \cos(2 \cdot \pi \cdot f \cdot t).$$

Comparing this with

$$x(t) = \cos(2 \cdot \pi \cdot 400 \cdot t),$$

we obtain

$$f = 400 \text{ Hz.}$$

(b)

The angular frequency is related to the regular frequency by

$$\omega = 2 \cdot \pi \cdot f.$$

Thus,

$$\omega = 2 \cdot \pi \cdot 400 = 800\pi \text{ rad/s.}$$

(c)

Using Euler's formula,

$$\cos(2 \cdot \pi \cdot 400 \cdot t) = \frac{1}{2}e^{j(2 \cdot \pi \cdot 400 \cdot t)} + \frac{1}{2}e^{-j(2 \cdot \pi \cdot 400 \cdot t)}.$$

Therefore, the Fourier transform with respect to f is

$$X(f) = \frac{1}{2}\delta(f - 400) + \frac{1}{2}\delta(f + 400).$$

(d)

According to the Nyquist-Shannon sampling theorem, the sampling frequency must satisfy

$$f_s \geq 2B.$$

Here, the highest frequency is

$$B = 400 \text{ Hz.}$$

Therefore, the minimum sampling frequency is

$$f_s = 2B = 800 \text{ Hz.}$$

(e)

Sampling at 600 Hz does not satisfy the Nyquist-Shannon sampling theorem because

$$600 < 800.$$

Therefore, the original signal cannot be perfectly reconstructed from samples taken at 600 Hz.

(f)

Aliasing occurs when the sampling frequency is too low. In that case, high-frequency components appear as lower-frequency components in the sampled signal. As a result, different continuous-time signals may produce the same discrete-time samples, and the original signal cannot be uniquely recovered.

5. We want to estimate the parameter vector

$$\boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}.$$

The prior estimate is

$$\hat{\boldsymbol{\theta}}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

and the prior covariance matrix is

$$M_0 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

A scalar measurement is obtained through

$$x_1 = \mathbf{h}^T \boldsymbol{\theta} + w_1,$$

where

$$\mathbf{h} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

and the measurement noise variance is

$$\sigma_w^2 = 1.$$

The observed value is

$$x_1 = 3.$$

- (a) Compute the Kalman gain vector K_1 .
- (b) Compute the updated estimate vector.
- (c) Compute the updated covariance matrix.
- (d) Explain the role of the Kalman gain in the update process.

Answer:

(a)

The Kalman gain is

$$K_1 = \frac{M_0 \mathbf{h}}{\sigma_w^2 + \mathbf{h}^T M_0 \mathbf{h}}.$$

First,

$$M_0 \mathbf{h} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Also,

$$\mathbf{h}^T \mathbf{M}_0 \mathbf{h} = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 4.$$

Therefore,

$$\mathbf{K}_1 = \frac{1}{1+4} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0.4 \\ 0.4 \end{bmatrix}.$$

(b)

The updated estimate is

$$\hat{\theta}_1 = \hat{\theta}_0 + \mathbf{K}_1 (\mathbf{x}_1 - \mathbf{h}^T \hat{\theta}_0).$$

Since

$$\mathbf{h}^T \hat{\theta}_0 = 0,$$

we obtain

$$\hat{\theta}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.4 \\ 0.4 \end{bmatrix} (3) = \begin{bmatrix} 1.2 \\ 1.2 \end{bmatrix}.$$

(c)

The updated covariance matrix is

$$\mathbf{M}_1 = (\mathbf{I} - \mathbf{K}_1 \mathbf{h}^T) \mathbf{M}_0.$$

First,

$$\mathbf{K}_1 \mathbf{h}^T = \begin{bmatrix} 0.4 \\ 0.4 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.4 & 0.4 \\ 0.4 & 0.4 \end{bmatrix}.$$

Therefore,

$$\mathbf{I} - \mathbf{K}_1 \mathbf{h}^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.4 & 0.4 \\ 0.4 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.4 \\ -0.4 & 0.6 \end{bmatrix}.$$

Thus,

$$\mathbf{M}_1 = \begin{bmatrix} 0.6 & -0.4 \\ -0.4 & 0.6 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1.2 & -0.8 \\ -0.8 & 1.2 \end{bmatrix}.$$

(d)

The Kalman gain controls how much the new measurement changes the prior estimate. A larger gain means that the estimate relies more strongly on the new observation, while a smaller gain means that the estimate relies more strongly on the prior estimate. The gain is determined by the relative size of the prior uncertainty and the measurement noise.